

How Externalities Affect Systems

IT TOOK ABOUT a billion years for life—single cell organisms—to develop on planet earth. Another 500 million years led to simple organisms that used photosynthesis; another 1–2 billion years later, multi-cellular life appeared. The next ~billion years led to plants, dinosaurs, birds, and mammals.

Evolution was the driving force of these changes. Suitability and adaptation was a key to competitive success. Life evolved so it could adapt to changing conditions. But every now and again, fate intervened, in the form of five mass extinctions.^{[200](#)}

The most recent: About 66 million years ago, a 7.5-mile-wide asteroid traveling about 50,000 mph slammed into the earth in the ocean near Chicxulub, Mexico. The collision resulted in an explosion estimated at over 100 trillion tons of TNT. That's a billion times more energy than the Hiroshima atomic bomb. The impact gouged the earth's crust several miles deep and more than 115 miles long.

It was not a good day to be a dinosaur.

Everything within 1,000 miles was killed by a giant fireball. That was followed by a 1,000-foot-high tsunami, extinguishing the fires, and drowning anything that survived. Just in case a few creatures were still standing, a 600-mile an hour wind blast tore through the region. And that was just day one. Nuclear winter followed, chilling the planet by a global average of 14 to 18 degrees. Eventually, the critters that could adapt to these circumstances did. The demise of the dinosaurs created an opportunity for the mammals.

What does any of this have to do with markets, economics, and investing? More than you might imagine.

The impact of externalities on economic systems is too often ignored.